Bochner - Kodaira inequality

Let $\alpha \in \Omega^{0,1}(L)$, $(L,h) \to (M,\omega)$
(M compact)
Ω = curvature of (L,h)

Then

$$\int_M (Ric_{j\bar{k}} + \Omega_{j\bar{k}})\, \alpha_{\bar{j}} \overline{\alpha_{\bar{k}}} \leq \|\bar{\partial}\alpha\|_{L^2}^2 + \|\bar{\partial}^*\alpha\|_{L^2}^2$$

where $\bar{\partial}^* : \Omega^{0,1}(L) \longrightarrow \Omega^0(L)$
$(= \Gamma(L))$

L^2 adjoint of $\underset{(=\nabla^{0,1})}{\bar{\partial}} : \Omega^0(L) \longrightarrow \Omega^{0,1}(L)$

$$\langle \bar{\partial}^*\alpha, \eta \rangle_{L^2} = \langle \alpha, \bar{\partial}\eta \rangle_{L^2}$$

in coordinates $\bar{\partial}^*\alpha = -g^{j\bar{k}} \nabla_j \alpha_{\bar{k}} = -\nabla_k \alpha_{\bar{k}}$

<u>Proof</u> in normal coords at a point

$$\bar{\partial}\alpha = \sum_{j,k} \nabla_{\bar{j}} \alpha_{\bar{k}}\, d\bar{z}^j \wedge d\bar{z}^k$$

$$= \sum_{j<k} (\nabla_{\bar{j}} \alpha_{\bar{k}} - \nabla_{\bar{k}} \alpha_{\bar{j}})\, d\bar{z}^j \wedge d\bar{z}^k$$

$$|\bar{\partial}\alpha|^2 = \sum_{j<k} \Big(\nabla_{\bar{j}} \alpha_{\bar{k}} \nabla_j \overline{\alpha_{\bar{k}}} + \nabla_{\bar{k}} \alpha_{\bar{j}} \nabla_k \overline{\alpha_{\bar{j}}} - \nabla_{\bar{j}} \alpha_{\bar{k}} \nabla_k \overline{\alpha_{\bar{j}}} - \nabla_{\bar{k}} \alpha_{\bar{j}} \nabla_j \overline{\alpha_{\bar{k}}} \Big)$$

$$= \sum_{j,k} \nabla_{\bar{j}} \alpha_{\bar{k}} \nabla_j \overline{\alpha_{\bar{k}}} - \nabla_{\bar{j}} \alpha_{\bar{k}} \nabla_k \overline{\alpha_{\bar{j}}}$$

$$|\bar\partial^*\alpha|^2 = \nabla_k \alpha_{\bar k} \nabla_{\bar j} \bar\alpha_{\bar j}$$

$$\int_M |\bar\partial\alpha|^2 + |\bar\partial^*\alpha|^2 = \int_M |\nabla^{0,1}\alpha|^2 + (\nabla_k \nabla_{\bar j} \alpha_{\bar k})\bar\alpha_{\bar j} - (\nabla_{\bar j}\nabla_k \alpha_{\bar k})\bar\alpha_{\bar j}$$

$$\geq \int_M \Big(\underbrace{R_{p\bar k k \bar j}\alpha_{\bar p}}_{Ric_{p\bar j}\alpha_{\bar p}} + \Omega_{k\bar j}\alpha_{\bar k}\Big)\bar\alpha_{\bar j}$$

$$= \int_M (Ric_{k\bar j} + \Omega_{k\bar j})\alpha_{\bar k}\bar\alpha_{\bar j} \qquad \square$$

<u>Theorem</u> (Hörmander) Let $(L,h) \to (M,\omega)$ compact s.t. $\sqrt{-1}(Ric_{k\bar j} + \Omega_{k\bar j}) \geq c\omega$, $c>0$.

If $\alpha \in \Omega^{0,1}(L)$ s.t. $\bar\partial\alpha = 0$, then $\exists\, f \in \Omega^0(L)$ s.t. $\bar\partial f = \alpha$ and

$$\int_M |f|_h^2 \leq \int_M c^{-1}|\alpha|_h^2$$

(in particular $H^{0,1}(M,L) = 0$)

<u>Proof</u> construct f by duality as linear functional $\lambda : L^2\Omega^0(L) \to \mathbb{C}$

<u>Note</u>: $\bar\partial f = \alpha$ means

$$\langle \bar\partial f, \eta\rangle_{L^2} = \langle \alpha, \eta\rangle_{L^2}$$
$$\parallel$$
$$\langle f, \bar\partial^*\eta\rangle_{L^2}$$

i.e. $\lambda(\bar\partial^* \eta) = \langle \alpha, \eta \rangle_{L^2}$

$\Rightarrow$ we define λ on $\operatorname{Im} \bar\partial^*$ (on smooth forms) by this formula.

<u>Claim</u>: $\bar\partial^* \eta \longmapsto \langle \alpha, \eta \rangle_{L^2}$

is bounded on $\operatorname{Im} \bar\partial^*$ by $\left(\int_M c^{-1}|\alpha|^2\right)^{1/2}$

i.e. $|\langle \alpha, \eta \rangle_{L^2}| \leq \left(\int_M c^{-1}|\alpha|^2\right)^{1/2} \cdot \|\bar\partial^* \eta\|_{L^2}$ $(*)$

Then can extend this functional to $L^2\Omega^0(L)$ by Hahn–Banach, with same norm bound.

<u>Proof of $(*)$</u> Write $\eta = \eta_1 + \eta_2$

under L^2-decomp into

$\eta_1 \in \ker \bar\partial$, $\eta_2 \perp \ker \bar\partial$

$\Rightarrow \bar\partial^* \eta_2 = 0$

Bochner–Kodaira $\Longrightarrow$

$$\left| \int_M \langle \alpha, \eta \rangle \right| = \left| \int_M \langle \alpha, \eta_1 \rangle \right| \leq \left(\int_M c^{-1}|\alpha|^2 \right)^{1/2} \left(\int_M c|\eta_1|^2 \right)^{1/2}$$

$$\leq \left(\int_M c^{-1}|\alpha|^2 \right)^{1/2} \left(\int_M (\mathrm{Ric}_{j\bar k} + \Omega_{j\bar k}) \eta_{1,j} \overline{\eta_{1,k}} \right)^{1/2}$$

$$\overset{(BK)}{\leq} \left(\int_M c^{-1}|\alpha|^2 \right)^{1/2} \left(\cancel{\|\bar\partial \eta_1\|^2} + \underset{\|\bar\partial^* \eta\|^2}{\underset{\shortparallel}{\|\bar\partial^* \eta_1\|^2}} \right)^{1/2}$$

$$= \left(\int_M c^{-1}|\alpha|^2 \right)^{1/2} \|\bar\partial^* \eta\|_{L^2} \qquad \square$$

<u>Theorem</u> (L^∞ & gradient bounds)

Let $(L,h) \to (M^n, \omega)$ s.t. $\sqrt{-1}\,\Omega = \omega$

$Ric(\omega) > -\omega$ and $Vol(B(p,1)) > v > 0$ $\forall p \in M$

$\exists\, C = C(n,v)$ s.t. if $s \in H^0(M,L)$, then

$$\sup_M \left(|s| + |\nabla s| \right) \leq C \|s\|_{L^2}$$

<u>Proof</u>

$$\begin{aligned}
\Delta |s|^2 &= \nabla_i \nabla_{\bar{i}} \langle s, s \rangle \\
&= |\nabla s|^2 + \langle s, \nabla_{\bar{i}} \nabla_i s \rangle \\
&= |\nabla s|^2 + \langle s, \nabla_i \nabla_{\bar{i}} s \rangle - \langle s, \Omega_{i\bar{i}} s \rangle \\
&\geq |\nabla s|^2 - n |s|^2
\end{aligned}$$

mean value inequality

$$\Rightarrow \quad |s|^2(p) \leq C \fint_{B(p,1)} |s|^2$$

and $\displaystyle\int_M |\nabla s|^2 = n \int_M |s|^2$

For ∇s we have

$$\begin{aligned}
\Delta |\nabla s|^2 &= \nabla_i \nabla_{\bar{i}} \langle \nabla_k s, \nabla_k s \rangle \\
&\geq \langle \nabla_i \nabla_{\bar{i}} \nabla_k s, \nabla_k s \rangle + \langle \nabla_k s, \nabla_{\bar{i}} \nabla_i \nabla_k s \rangle \\
&= \langle \nabla_i \nabla_k \nabla_{\bar{i}} s - \nabla_i (\Omega_{k\bar{i}} s), \nabla_k s \rangle + \langle \nabla_k s, \nabla_i \nabla_{\bar{i}} \nabla_k s \\
&\qquad + R_{k\bar{p}i\bar{i}} \nabla_p s \\
&\qquad - \Omega_{i\bar{i}} \nabla_k s \rangle
\end{aligned}$$

$$= -|\nabla s|^2 + \langle \nabla_k s, \cancel{\nabla_i \nabla_k \nabla_{\bar i} s} + \nabla_i(\Omega_{k\bar i} s) + Ric_{k\bar p} \nabla_p s - \underline{n \nabla_k s} \rangle$$

$$\geq -(n+1)|\nabla s|^2 + |\nabla s|^2 - |\nabla s|^2$$

$$\Delta |\nabla s|^2 \geq -(n+1)|\nabla s|^2$$

mean value ineq $\Rightarrow$

$$|\nabla s|^2(p) \leq C \fint_{B(p,1)} |\nabla s|^2 \leq C \int_M |s|^2 \quad \square$$

Constructing non-vanishing sections

<u>Thm</u> (Donaldson-Sun) Let $v, \delta > 0$.

$\exists\, K, \varepsilon, C > 0$ dep. n, δ, v satisfying the following. Suppose $(L, h) \to (M, \omega)$, $\sqrt{-1}\Omega = \omega$, $Ric(\omega) > -\varepsilon\omega$, $Vol\, B(q,1) > v$, $\forall q \in M$.

Suppose $d_{GH}(B(p, \varepsilon^{-1}), B(0, \varepsilon^{-1})) < \varepsilon$

for some $p \in M$, $0 \in C(X)$ cone.

Then $\exists\, m < K$ and section $s \in H^0(M, L^m)$ s.t. $\|s\|_{L^2} < C$ and $\left| |s(x)| - e^{-\frac{m}{2} dist(p,x)^2} \right| < \delta$

$\forall x \in M$